



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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TECHNICA PRESENTATION

CSA5112 CRYPTOGRAPHY AND NETWORK SECURITY

**TOPIC: SUBSTITUTIONAL VS TRANSPOSITION TECHNIQUE
WHICH PROVIDES BETTER SECURITY**

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INTRODUCTION

What is Classical Encryption?

Classical encryption techniques protect confidential information by transforming plaintext into ciphertext.

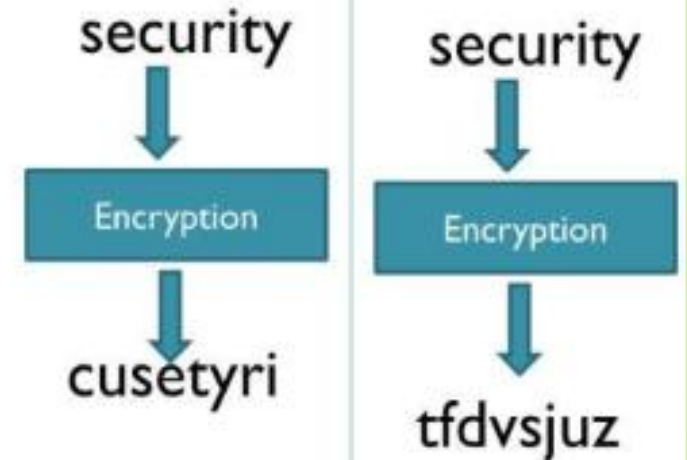
Two fundamental techniques are:

- **Substitution Technique**

Replaces each character with another character or symbol.

- **Transposition Technique**

Rearranges the positions of characters without changing them.



Substitution Technique

What is Substitution?

Substitution encryption replaces each plaintext character with another character according to a fixed rule or secret key. The positions of the characters remain unchanged.

How it Works

Plaintext: HELLO

Using a Caesar Cipher (key= 3):

Encrypted Text: KHOOR

$$C = (P + K) \bmod 26$$

$$P = (C - K) \bmod 26$$

Plaintext	H	E	L	L	O
Ciphertext	K	H	O	O	R

EXAMPLES

Monoalphabetic Cipher

Playfair Cipher

Vigenère Cipher.

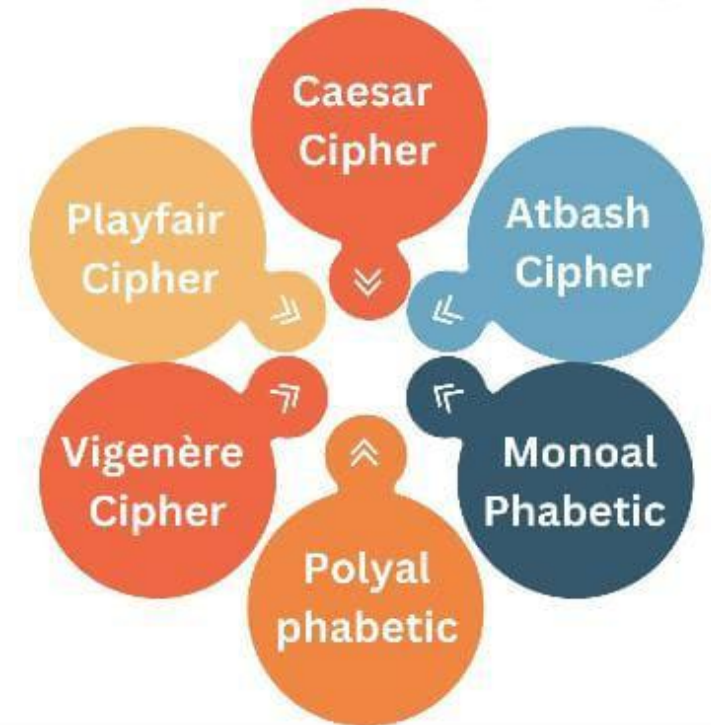
Hill Cipher

Advantages

- ✓ Hides the original characters (Confusion)
- ✓ Easy to implement
- ✓ More secure with complex substitution methods

Limitations

- ✗ Simple substitution ciphers are vulnerable to frequency analysis.



TRANSPOSITION TECHNIQUE

What is Transposition?

Transposition encryption does not change the characters. Instead, it rearranges their positions using a secret pattern or key.

How it Works

Plaintext: HELLO

Rearranging the positions:

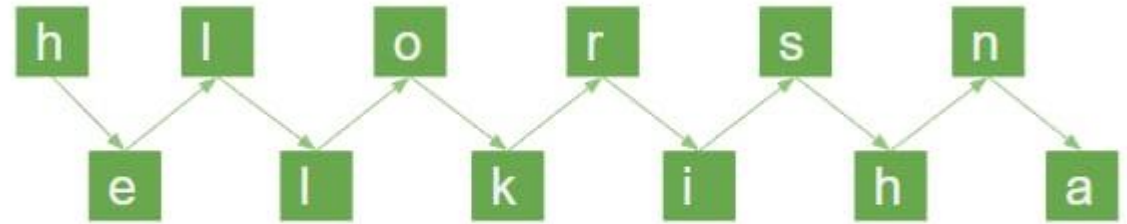
Reordered:

Ciphertext: HLOEL

Original Position	1	2	3	4	5
Letter	H	E	L	L	O

Examples

- ▶ Rail Fence Cipher
- ▶ Columnar Transposition



Advantages

- ▶ ✓ Hides the message structure (Diffusion)
- ▶ ✓ Difficult to interpret without the correct key
- ▶ ✓ More effective when combined with substitution

Limitations

- ▶ ✗ Character frequency remains unchanged, making it vulnerable to certain attacks if used alone.

Security Analysis

Substitution vs Transposition

Security Aspect	Substitution Technique	Transposition Technique
Encryption Method	Replaces characters	Rearranges character positions
Security Principle	Confusion	Diffusion
Protects	Character identity	Character arrangement
Resistance to Pattern Recognition	High	Moderate
Resistance to Frequency Analysis	Better (especially advanced substitution ciphers)	Lower (letter frequencies remain unchanged)
Difficulty to Break	Higher	Lower
Standalone Security		
Overall Verdict	 Better Security	Moderate Security

CONCLUSION

- ▶ Substitution and Transposition are two fundamental classical encryption techniques.
- ▶ Substitution secures data by replacing characters, while Transposition secures data by rearranging their positions.
- ▶ Based on the security analysis, Substitution provides better standalone security because it hides the actual identity of the characters, making the ciphertext more difficult to interpret.
- ▶ However, neither technique alone is sufficient against modern cryptographic attacks.
- ▶ Modern encryption algorithms, such as DES and AES, combine substitution and transposition to achieve stronger security through both confusion and diffusion.